
PHYSIGYM: Bridging Agent-Based Biological Simulation and Reinforcement Learning A State-Space Study on Tumor Microenvironment Control

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Abstract

1 This paper presents PHYSIGYM, a framework that integrates agent-based biological
2 simulation within standardised reinforcement learning environments. By connect-
3 ing the agent-based modelling framework PHYSICELL with the Gymnasium API,
4 we provide a flexible tool for exploring RL strategies that control *in silico* biolog-
5 ical processes. We demonstrate PHYSIGYM’s potential through a case study on
6 a simplified tumor microenvironment (TME) toy model, in which a Soft Actor-
7 Critic agent learns targeted drug-delivery policies that steer the microenvironment
8 toward an anti-tumoral state and, in the best episodes, achieve tumor elimination.
9 We further use this case study to systematically compare a family of observation
10 spaces from a partially observed macrophage-count baseline, through spatial scalar
11 summaries, to full multi-channel spatial images. Training on a structured rectangle
12 layout and testing on a held-out network-field layout, we find that spatially ex-
13 plicit image observations generalize markedly better than scalar summaries: image
14 modes reach a mean held-out test return of $\approx +30$ to $+32$, whereas spatial-scalar
15 summaries train comparably ($+43$) but collapse to ≈ 0 out of distribution, and the
16 macrophage-only POMDP baseline fails to learn at all (-32). This demonstrates
17 that PHYSIGYM’s flexible state-space interface directly impacts policy quality and
18 generalization in spatially structured environments. PHYSIGYM is open-source at
19 <https://anonymous.4open.science/r/PhysiGym-76A7>.

20 1 Introduction

21 Biological systems are complex, dynamic, and often difficult to study in real life. In silico modeling
22 aims to mimic these systems, providing a controlled environment for exploring specific aspects of
23 biological processes and testing hypotheses. In particular, agent-based models (ABMs) provide
24 a powerful framework for simulating biological systems where agents (e.g. cells) interact dynam-
25 ically based on predefined rules. These models are used in oncology and immunology to study
26 emerging behaviors in complex biological systems, with the potential to propose novel therapeutic
27 strategies [Johnson et al., 2025, Laubenbacher et al., 2024, Rocha et al., 2024].

28 Ordinary differential equations (ODEs) and agent-based models (ABMs) are widely used to simulate
29 the tumor microenvironment (TME). ODEs provide compact system-level descriptions, while ABMs
30 capture spatial and cellular heterogeneity at higher computational cost. For dynamic treatment
31 regimes (DTRs) [Gray and Pallavicini, 1982], however, simulation alone is insufficient: designing
32 adaptive, sequential interventions requires a control framework. Reinforcement learning (RL), where
33 an RL agent learns policies through interaction with an environment, offers such a framework [Barto
34 et al., 1989]. By modeling treatment regimes as a sequence of decisions, RL enables adaptive control

of biological simulations [Chakraborty and Murphy, 2014]. For efficient experimentation and policy learning, RL further requires a standardized interface. To address this, we introduce PHYSIGYM, an interface that connects PHYSICELL [Ghaffarizadeh et al., 2018], an ABM framework for multicellular simulations of tissues or cell cultures (tumor microenvironment, other disease tissues, or bacteria colonies), with Gymnasium [Towers et al., 2024], a widely adopted RL application programming interface (API). PHYSICELL enables flexible multiscale modeling of biological systems, coupled with a sophisticated underlying physics simulator (the BioFVM diffusion transport solver [Ghaffarizadeh et al., 2016]), while Gymnasium provides a structured framework for training and evaluating RL policies.

The first implementation of RL on an ABM framework based on BioFVM was proposed by Zade et al. [2020], which applied Q-learning [Watkins and Dayan, 1992] to optimize Temozolomide treatment regimes for patients with glioblastoma multiforme, a highly specific problem. More recently, Aif et al. [2025] used PHYSICELL as a complex (high-fidelity) simulation environment for tumor growth, combined with a simple (low-fidelity) RL training environment of coupled ordinary differential equations (ODE) to learn treatment strategies for solid tumors.

PHYSIGYM bridges PHYSICELL and Gymnasium to provide a reproducible and scalable framework for integrating agent-based simulations of biological systems with RL strategies. In the following, we present PHYSIGYM’s motivation and design, demonstrating how it directly connects ABMs and RL to control complex biological systems. In particular, we focus on the tumor microenvironment, a biological system made up of cancer cells and surrounding immune and stromal cells that engage in complex interactions and cross-talks, which is proven to be crucial for an effective response to immunotherapy, an important therapeutic approach in an increasing number of types of cancer.

To illustrate the framework, we build a simplified TME toy model and use it as an end-to-end RL case study. This model also serves as a controlled testbed for a practical question that arises in any spatially explicit biological environment: among the many observation spaces PHYSIGYM can expose (scalars, images, graphs), which ones help an RL agent learn an effective dynamic treatment regime? We find that image-based observations that preserve the spatial layout of cells and substrates outperform compact scalar summaries, both in cumulative return and in generalization to held-out spatial configurations. This is an application-level finding on a deliberately simplified model; the primary contribution of this work is PHYSIGYM itself.

Contributions. (1) PHYSIGYM: an open-source framework that bridges the PHYSICELL ABM and the Gymnasium RL API, turning any PHYSICELL model into an RL environment with minimal code changes. (2) A simplified TME toy model demonstrating an end-to-end RL case study in which an agent learns a spatial drug-delivery policy that drives the microenvironment toward an anti-tumoral state. (3) A systematic comparison of observation spaces on the toy model, showing that spatially explicit (image) representations help RL learn and generalize better than scalar summaries.

2 Methods

2.1 PHYSIGYM design and implementation

PHYSICELL is an open-source ABM framework written in C++ that models multicellular systems using classical mechanics and the BioFVM diffusive transport solver [Ghaffarizadeh et al., 2018, 2016]. Cells are agents governed by type-specific rule sets; substrates such as oxygen or cytokines diffuse through the domain. Intracellular signalling networks [Letort et al., 2018, Ponce-de Leon et al., 2023, Ruscone et al., 2024] and extracellular matrix fibres [Noël et al., 2024] can be added, yielding ABMs that are spatially explicit in 2D or 3D, off-lattice, and multiscale [Metzcar et al., 2019].

RL frames sequential decision-making as a Markov Decision Process (MDP) [Bellman, 1957, Puterman, 2014] with state space S , action space A , transition model T , and reward R . The objective is

$$\arg \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=0}^{\tau} \gamma^t r_t \mid s_0, \pi \right].$$

Gymnasium [Towers et al., 2024] is the standard Python interface for MDPs, promoting modularity and code reuse across algorithms and environments.

PHYSIGYM connects these two ecosystems by extending the Python interpreter [van Rossum, 1995, Python Software Foundation, 2024] with PHYSICELL. The C++ simulation loop is split into three Python-callable functions: `physicell_start`, `physicell_step`, and `physicell_stop`. For most models only `physicell_step` needs to be modified (e.g. to inject a drug). Three read functions `get_cell`, `get_microenv`, and `get_graph` return cell positions, substrate concentration fields, and cell neighbourhood graphs respectively; all are implemented in `physicellmodule.cpp`.

The interface between PHYSICELL and Gymnasium uses PHYSICELL’s built-in *user parameter* and *custom data* constructs, configurable through PhysiCell Studio [Heiland et al., 2024] without recompilation. Users subclass `CorePhysiCellEnv` in `physicell_model.py` to specify observation space, action space, and reward; the base class handles all model-independent Gymnasium bookkeeping. Any existing PHYSICELL model can therefore become a PHYSIGYM environment with minimal code changes. PHYSIGYM is tested on Linux, WSL, and macOS, with continuous integration via GitHub Actions. Documentation and tutorials are at <https://anonymous.4open.science/r/PhysiGym-76A7/man>.

2.2 Implementation of a Tumor Microenvironment Toy Model

We developed a simplified tumor microenvironment (TME) *toy model* to illustrate the potential uses of PHYSIGYM. The model is deliberately small and computationally cheap so that we can train RL agents to convergence many times over (across observation spaces and seeds), while still retaining the spatial, multicellular, and chemically coupled structure that distinguishes agent-based models (ABMs) from ODE surrogates. It includes three cell types, five diffusible substrates, and a set of simple agent-based rules. This toy model highlights how the RL framework can control tumor progression by modulating the whole microenvironment, rather than solely targeting cancer-cell killing.

Domain and Dynamics. The continuous-space environment is simulated on a $64 \times 64 \mu\text{m}$ 2D domain. The RL gym step interval is $\Delta t = 15$ min, with episodes running up to 10,080 min (7 simulated days).

Entity Dynamics & The Control Challenge. The environment formulates a highly non-linear, indirect control problem. The agent’s action (deploying a localized therapeutic, `drug_1`) has zero direct cytotoxicity. Instead, the agent must learn to manipulate the environment’s underlying transition dynamics by reversing a spatial immunosuppression loop. The system comprises three interacting cellular populations whose movements are governed by biochemical spatial gradients (chemotaxis):

- **Tumor Cells (The Target):** Immobile entities that continuously divide. They actively modify the environment by secreting a chemical signal (`tumor_molecule`) that suppresses the local immune system.
- **Macrophages (The Mediators):** Mobile, immortal entities that navigate toward the tumor. They exhibit state-dependent behavior: the tumor’s chemical signal successfully “hijacks” them, transforming them from an anti-tumoral to a pro-tumoral state.
- **T-cells (The Effectors):** Mobile, immortal entities capable of triggering tumor apoptosis. However, their spatial navigation is strongly repelled by the presence of pro-tumoral macrophages.

This setup creates a hostile, self-reinforcing feedback loop: the tumor protects itself by converting macrophages, which in turn dynamically block T-cells from reaching the target. As illustrated in Figure 1, the RL agent must maximize reward by learning to precisely deploy its drug to “re-program” the macrophages back to an anti-tumoral state. This spatial intervention remodels the chemical gradients, clearing a path for T-cells to navigate to the tumor and execute their killing function.

2.3 Implementation of an RL approach to control the TME

Our overall objective is to achieve complete tumor eradication while minimizing drug exposure. We describe the action space A , the reward R , and the training and test distributions. The observation space S is the component we vary across experiments: in Section 3 we compare a family of observation modes spanning a partially observed scalar baseline, spatial scalar summaries, and

information, others are spaceless summaries (e.g. a small vector of global statistics). Because PHYSI-GYM can build all of these from the same underlying simulation, we use the toy model to study how the choice of observation space affects the dynamic treatment regime that RL learns, comparing nine modes on cumulative return and on generalization to unseen spatial configurations.

We compare observation modes spanning a spectrum from a partially observed scalar baseline to full multi-channel spatial images. Let $N_c = 3$ (tumor, T-cell, macrophage) and $N_s = 5$ substrates. Several modes optionally split the macrophage population into its *M1* (anti-tumoral) and *M2* (pro-tumoral) polarisation states, which we denote with the suffix *m1m2*; the classification rule is given below.

Macrophage polarisation (M1/M2). Macrophages do not carry an explicit polarisation label in the simulator; their state is implicit in the local chemical environment. We recover it from the microenvironment: for each alive macrophage we look up the two competing substrate fields at its nearest voxel and classify it as *M2* (pro-tumoral) if $\text{conc}_{\text{pro_tumoral}} > \text{conc}_{\text{anti_tumoral}}$ at that voxel, and *M1* (anti-tumoral) otherwise. This makes the pro-/anti-tumoral balance the very quantity the agent must flip observable as a count (scalars modes) or as two spatial channels (*img* modes), rather than leaving it latent in the substrate fields.

Scalar Observation Spaces

scalars_macrophages (POMDP baseline): Alive count per cell type, with the macrophage slot split into M1 and M2 counts; each is normalised to the initial population ($s = c/c_{\text{init}} - 1$). This mode exposes only *how many* macrophages are polarised each way and carries no spatial information at all neither cell positions nor substrate maps. It is therefore a strongly partially observed (POMDP) baseline: the true state that governs the transition dynamics (where cells and gradients are) is hidden, and the agent must act on an aggressively compressed summary.
Space: $\mathbb{R}^{N_c+1}$

spatial_scalars_cells_substrates: Per-type normalised cell counts and per-substrate maxima, augmented with 6 *k*-means spatial statistics per cell type (cluster occupancy and centroids), giving a compact spatial summary without full images.
Space: $\mathbb{R}^{N_c+N_s+6kN_c}$

spatial_scalars_cells_m1m2 / ..._substrates_m1m2: As above but with the macrophage population split into M1/M2 for both the scalar counts and the *k*-means spatial features (*..._substrates_m1m2* additionally keeps the substrate maxima).
Space: $\mathbb{R}^{(N_c+1)+[N_s]+6k(N_c+1)}$

spatial_scalars_cells_spatial_no_scalars_substrates_m1m2: M1/M2 cell counts plus *k*-means spatial features for *both* cell types and substrate fields (spatially resolved substrate summaries in place of scalar maxima).
Space: $\mathbb{R}^{(N_c+1)+6kN_s+6k(N_c+1)}$

Image-Based Observation Spaces

img_mc_cells (I1): One 64×64 channel per cell type.
Space: $\mathbb{R}^{N_c \times 64 \times 64}$ (uint8)

img_mc_cells_m1m2: I1 with the macrophage channel split into separate M1 and M2 spatial maps.
Space: $\mathbb{R}^{(N_c+1) \times 64 \times 64}$ (uint8)

img_mc_cells_substrates (I2): I1 concatenated with all five substrate concentration maps, providing direct spatial feedback on drug distribution and the polarising factors.
Space: $\mathbb{R}^{(N_c+N_s) \times 64 \times 64}$ (uint8)

img_mc_cells_substrates_m1m2 (full information): I2 with the additional M1/M2 macrophage channels. This mode carries every signal the others expose cell positions, substrate fields,

210 and the explicit polarisation split and serves as the near-fully-observed upper reference against which
 211 the compressed modes are compared.
 212 $Space: \mathbb{R}^{(N_c+N_s+2) \times 64 \times 64}$ (uint8)

213 An observability spectrum

214 Together these modes trace an observability spectrum, from `scalars_macrophages` a POMDP that
 215 sees only polarisation counts to `img_mc_cells_substrates_m1m2`, which spatially resolves every
 216 state variable relevant to the control problem. The indirect drug mechanism (Section 2.2) requires
 217 multi-step credit assignment: the agent injects drug, macrophages repolarise over several steps,
 218 anti-tumoral gradients shift, T-cells migrate, and only then do tumor cells die. Scalar summaries
 219 cannot tell the agent whether an injection reached macrophages, whether polarisation occurred, or
 220 where T-cells will concentrate next; only the per-pixel substrate and M1/M2 channels of the image
 221 modes make the polarisation state and drug distribution spatially legible at each step.

222 3 Results

223 3.1 Aggregated performance

224 Table 1 summarises final performance (seeds averaged). Figures 2 and 4 show learning curves vs.
 225 cumulative environment steps with mean $\pm$ std shading across seeds.

Table 1: Performance at end of training (train = rectangle, test = network-field, seeds averaged). $Train_\mu/Test_\mu$: mean EWMA-50 return. $Train_\sigma/Test_\sigma$: std across seeds; n = number of seeds. This is the curated main set; the remaining m1m2 scalar variants appear in Appendix C, and 95% bootstrap confidence intervals on the mean in Appendix D. RAND is a random-action policy (5 seeds), included as a learning-free reference: modes below it fail to beat random behaviour.

ID	Mode	n	$Train_\mu$	$Train_\sigma$	$Test_\mu$	$Test_\sigma$
I2m	<code>img_mc_cells_substrates_m1m2</code>	5	+40.8	8.5	+32.5	9.4
I1	<code>img_mc_cells</code>	6	+30.5	8.7	+32.4	10.5
I2	<code>img_mc_cells_substrates</code>	7	+42.5	6.6	+30.4	7.4
S3s	<code>spatial_scalars_cells_substrates</code>	5	+42.9	18.8	−0.0	9.1
RAND	random policy (baseline)	5	−18.4	4.0	−25.3	3.0
POMDP	<code>scalars_macrophages</code>	5	−20.4	8.3	−31.8	2.3

226 3.2 Key findings

227 **Generalization, not training fit, separates the modes.** On the training rectangle layout the spatial-
 228 scalar summary S3s is fully competitive its training return (+42.9) matches the best image mode I2
 229 (+42.5). The distinction appears only out of distribution: on the held-out network-field, the three
 230 image modes retain $\approx +30$ to $+32$ test return, while S3s collapses to -0.0 . Training fit alone would
 231 have ranked the scalar summary among the best; only the OOD test reveals that it has overfit the
 232 fixed strip geometry of the rectangle. This separation is statistically supported: the image modes’
 233 95% bootstrap confidence intervals on the mean test return lie entirely above $+24$ and do not overlap
 234 those of any spatial-scalar mode, all of which contain zero (Appendix D).

235 **Spatially explicit observations transfer; scalar summaries do not.** All three image modes
 236 `img_mc_cells` (I1, +32.4), `img_mc_cells_substrates` (I2, +30.4), and the full-information
 237 `img_mc_cells_substrates_m1m2` (I2m, +32.5) generalize to the network-field, and the gaps
 238 among them are within one seed-standard-deviation. That I1 already generalizes well indicates that,
 239 in this toy model, preserving the *spatial layout* of cells is the primary driver of transfer; the extra
 240 substrate and M1/M2 channels of I2/I2m do not further separate on average, though I2m gives the
 241 single best test return.

242 **The POMDP baseline is no better than acting randomly.** `scalars_macrophages`, which
 243 observes only M1/M2 counts and no spatial information, fails to improve on either distribution

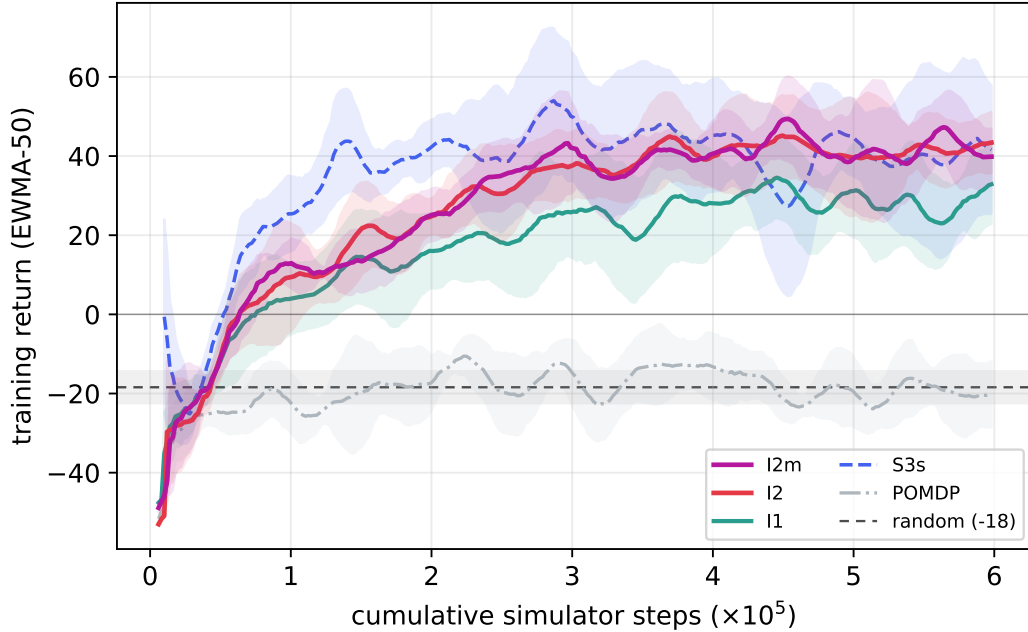


Figure 2: Training return on the rectangle layout (EWMA-50, mean $\pm 1\sigma$ across seeds) vs. cumulative *simulator* steps ($\times 10^5$; each of the 10^5 agent decisions is held for 6 simulator steps, giving a 6×10^5 budget). All modes learn on the training layout except the scalars_macrophages POMDP baseline, which flatlines near -20 . Spatial-scalar summaries (S3s) train as fast as, or faster than, the image modes here the separation only appears out of distribution (Figure 4).

(train -20.4 , test -31.8) and, crucially, does *not* beat a random-action policy (train -18.4 , test -25.3 ; RAND in Table 1). Its test return is if anything below random, and the two bootstrap CIs barely overlap (Appendix D): the macrophage-count signal is not merely uninformative but actively misleading, steering dosing worse than chance. Knowing *how many* macrophages are polarised without knowing *where* the drug, cells, and gradients are is insufficient to solve the indirect, spatially mediated control problem a concrete illustration of how observation-space choice can render an otherwise learnable task effectively unlearnable. By contrast, every spatial-scalar mode, though it fails to generalize, still clears the random baseline on test (≥ 0 vs. -25.3), confirming they learn *some* transferable structure just far less than the image modes.

On seed robustness. Unlike a cleaner earlier run, we do not observe a systematic seed-variance advantage for any family on this benchmark: end-of-training test std is 7–12 for the image modes and 9 for S3s (Table 1, Figure 3). The POMDP baseline’s low std (2.3) reflects consistent failure rather than robustness.

3.3 The role of the M1/M2 split and spatial scalars

The full mode family (Appendix C, Table 3) lets us isolate the effect of the explicit M1/M2 polarisation split. Adding it to the cell image (I1 $\rightarrow$ img_mc_cells_m1m2) does not improve average OOD test return ($+32.4 \rightarrow +26.0$), and adding it on top of the full image (I2 $\rightarrow$ I2m) leaves it essentially unchanged ($+30.4 \rightarrow +32.5$): once the substrate fields are spatially available, the CNN can already read out polarisation from the pro_tumoral/anti_tumoral channels, so the pre-computed split is redundant. Among the spatial-scalar modes, none generalize (all within $[-0.0, +9.5]$ test return) regardless of whether they carry the M1/M2 split or spatially resolved substrate summaries the common failure is the loss of per-cell spatial layout, which the k -means summaries do not preserve.

Appendix E shows a matched example episode comparing an image mode, a spatial-scalar mode, and the POMDP baseline rolled out from an identical network-field initial condition, illustrating the failure mode that summarises the comparison: without a spatially explicit representation the agent

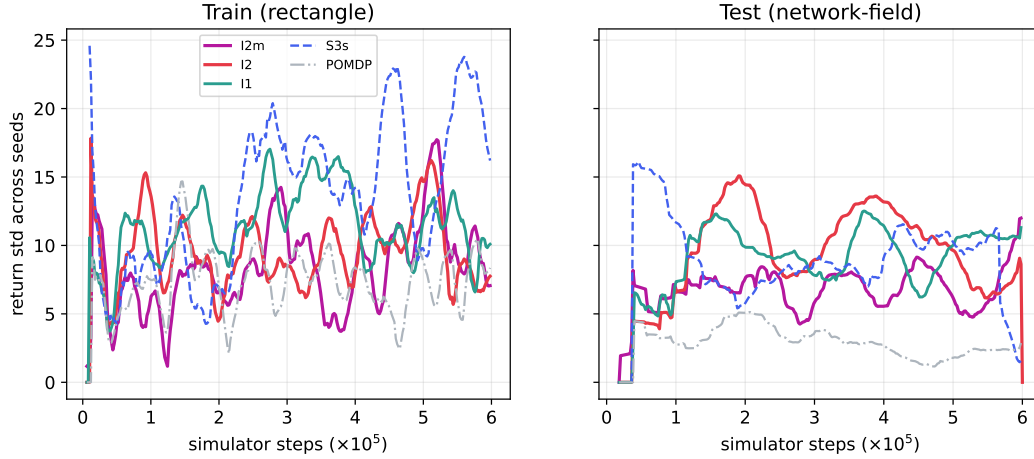


Figure 3: Standard deviation of return across seeds vs. simulator steps: training on rectangle (left) and test on network-field (right). Inter-seed variability is broadly comparable across the learning modes and fluctuates over training, so we do not claim a systematic variance advantage for any single family; the POMDP baseline has the lowest test variance simply because it converges to a uniformly poor policy. At end of training the image modes reach test return $\approx +30$ with std $\approx 7\text{--}12$ (Table 1).

cannot target the drug precisely enough to convert dose into effective macrophage repolarisation on an unseen layout the image mode reaches near-eradication (3 tumor cells) while the scalar and POMDP modes leave 70 and 130 cells respectively.

4 Discussion

The central outcome of this work is PHYSIGYM itself: a framework that lets RL agents control a full agent-based TME through a standardised interface, without recompilation, and with arbitrary observation and action spaces. The toy-model case study shows the framework end-to-end an agent learns a spatial drug-delivery policy that steers the microenvironment toward an anti-tumoral state and, in the best episodes, eradicates the tumor.

The toy model also produces a clear result on observation spaces, and crucially one that only surfaces under distribution shift. On the training rectangle layout, spatial-scalar summaries are fully competitive with image observations; it is on the held-out network-field that spatially explicit (image) observations pull ahead, generalizing to $\approx +30$ test return while the scalar summaries collapse to ≈ 0 . The CNN in the image modes can compute cross-channel co-localisation statistics (e.g. whether drug_1 and macrophages overlap at the same location) directly from the pixel layout statistics that transfer across geometries whereas scalar modes that encode k -means cluster centroids and absolute positions overfit the fixed strip geometry of the training layout. Notably, once cell layout is spatially available, preserving the substrate fields (I2) or pre-computing the M1/M2 split (I2m) does not further improve average transfer over the cell-only image (I1) on this toy model, though the full-information mode gives the single best test return.

We emphasise that this is an application-level observation on a deliberately simplified model, not a universal claim about biological control. In more complex or realistic models or with different reward designs, action spaces, or RL algorithms the relative value of cell versus substrate, and scalar versus image, may differ. PHYSIGYM is precisely the tool that makes such questions answerable in a reproducible way.

Limitations. The toy model is intentionally small ($64\mu\text{m}$, 2D) and cheap to simulate. Scaling to 3D or larger domains will increase the cost of image observations. All experiments use a single RL algorithm (SAC); alternative algorithms or model-based methods may interact differently with observation spaces. Each mode is trained with a modest number of seeds ($n = 5\text{--}7$), unevenly distributed across modes; we mitigate this with bootstrap confidence intervals (Appendix D) rather than relying on raw seed-standard-deviation, but the conclusions would be strengthened by more

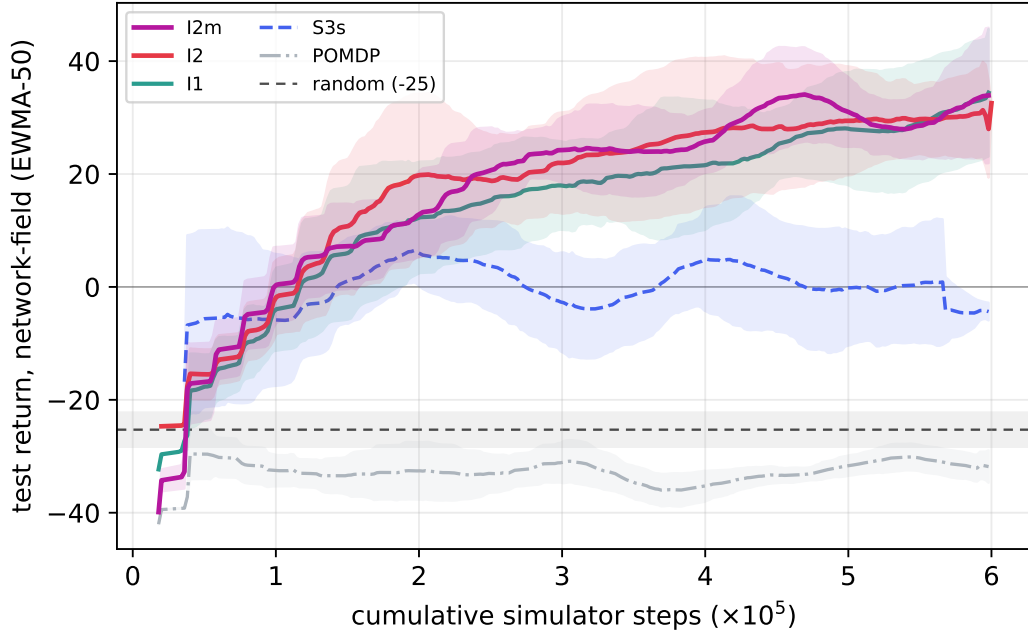


Figure 4: Held-out test return on the network-field layout (mean $\pm 1\sigma$ across seeds, EWMA-50) vs. cumulative simulator steps. The image modes (I2m, I2, I1) climb steadily to $\approx +30$, whereas the spatial-scalar summary (S3s) which trained *best* (Figure 2) collapses to ≈ 0 , revealing that it overfits the fixed rectangle geometry rather than learning transferable structure. The dashed line marks the random-policy baseline (test ≈ -25 , shaded $\pm 1\sigma$); the `scalars_macrophages` POMDP baseline tracks at or below it throughout, i.e. never beats random. Generalization, not training fit, is where spatially explicit observations separate from scalar summaries.

seeds and by testing transfer in both directions between layouts. We also study a single train/test layout pair (rectangle $\rightarrow$ network-field); the reverse direction is left to future work. Complex biological models and/or 3D models are left to future work.

5 Conclusion

We presented PHYSIGYM, an open-source framework that bridges the PHYSICELL ABM with Gymnasium RL, enabling direct policy learning in spatially explicit biological simulations. Using a TME model with an indirect drug mechanism, we showed that spatially explicit image observations are the critical state representation for *generalization*: training on a structured rectangle layout and testing on a held-out network-field, image modes retain $\approx +30$ to $+32$ test return while spatial-scalar summaries competitive during training collapse to ≈ 0 out of distribution, and a macrophage-count-only POMDP baseline fails to learn at all. The gap is a generalization phenomenon: what separates the representations is not training fit but whether the encoded structure transfers across spatial configurations. PHYSIGYM provides the infrastructure to study such questions in a principled, reproducible manner, and we anticipate it will serve as a testbed for both computational biology and RL communities.

References

- S. Aif, M. Eiche, N. Appold, E. Fischer, T. Citak, and J. Kayser. Reinforcement failing guides the discovery of emergent spatial dynamics in adaptive tumor therapy. Apr. 2025. doi: 10.1101/2025.04.08.647768.
- A. G. Barto, R. S. Sutton, and C. Watkins. *Learning and sequential decision making*, volume 89. University of Massachusetts Amherst, MA, 1989.

321 R. Bellman. A markovian decision process. *Journal of mathematics and mechanics*, pages 679–684,
322 1957.

323 B. Chakraborty and S. A. Murphy. Dynamic treatment regimes. *Annual review of statistics and its
324 application*, 1(1):447–464, 2014.

325 L. Espeholt, H. Soyer, R. Munos, K. Simonyan, V. Mnih, T. Ward, Y. Doron, V. Firoiu, T. Harley,
326 I. Dunning, et al. Impala: Scalable distributed deep-rl with importance weighted actor-learner
327 architectures. In *International conference on machine learning*, pages 1407–1416. PMLR, 2018.

328 A. Ghaffarizadeh, S. H. Friedman, and P. Macklin. Biofvm: an efficient, parallelized diffusive
329 transport solver for 3-d biological simulations. *Bioinformatics*, 32(8):1256–1258, 2016.

330 A. Ghaffarizadeh, R. Heiland, S. H. Friedman, S. M. Mumenthaler, and P. Macklin. PhysiCell: An
331 openb source physics-based cell simulator for 3-d multicellular systems. *PLOS Computational
332 Biology*, 14(2):e1005991, feb 2018. doi: 10.1371/journal.pcbi.1005991.

333 J. W. Gray and M. G. Pallavicini. Ara-c scheduling: Theoretical and experimental considerations.
334 *Medical and Pediatric Oncology*, 10(S1):93–108, 1982. ISSN 1096-911X. doi: 10.1002/mpo.
335 2950100711.

336 T. Haarnoja, A. Zhou, K. Hartikainen, G. Tucker, S. Ha, J. Tan, V. Kumar, H. Zhu, A. Gupta,
337 P. Abbeel, et al. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*,
338 2018.

339 R. Heiland, D. Bergman, B. Lyons, G. Waldow, J. Cass, H. L. da Rocha, M. Ruscone, V. Noël, and
340 P. Macklin. Physicell studio: a graphical tool to make agent-based modeling more accessible.
341 *Gigabyte*, 2024:gigabyte128, 2024.

342 J. A. Johnson, D. R. Bergman, H. L. Rocha, D. L. Zhou, E. Cramer, I. C. Mclean, Y. W. Dance,
343 M. Booth, Z. Nicholas, T. Lopez-Vidal, A. Deshpande, R. Heiland, E. Bucher, F. Shojaeian,
344 M. Dunworth, A. Forjaz, M. Getz, I. Godet, F. Kurtoglu, M. Lyman, J. Metzcar, J. T. Mitchell,
345 A. Raddatz, J. Solorzano, A. Sundus, Y. Wang, D. G. DeNardo, A. J. Ewald, D. M. Gilkes, L. T.
346 Kagohara, A. L. Kiemen, E. D. Thompson, D. Wirtz, L. D. Wood, P.-H. Wu, N. Zaidi, L. Zheng,
347 J. W. Zimmerman, J. M. Phillip, E. M. Jaffee, J. W. Gray, L. M. Coussens, Y. H. Chang, L. M.
348 Heiser, G. L. Stein-O’Brien, E. J. Fertig, and P. Macklin. Human interpretable grammar encodes
349 multicellular systems biology models to democratize virtual cell laboratories. *Cell*, July 2025.
350 ISSN 0092-8674. doi: 10.1016/j.cell.2025.06.048.

351 R. Laubenbacher, F. Adler, G. An, F. Castiglione, S. Eubank, L. L. Fonseca, J. Glazier, T. Helikar,
352 M. Jett-Tilton, D. Kirschner, et al. Toward mechanistic medical digital twins: some use cases in
353 immunology. *Frontiers in Digital Health*, 6:1349595, 2024.

354 G. Letort, A. Montagud, G. Stoll, R. Heiland, E. Barillot, P. Macklin, A. Zinovyev, and L. Calzone.
355 PhysiBoSS: a multi-scale agent-based modelling framework integrating physical dimension and
356 cell signalling. *Bioinformatics*, 35(7):1188–1196, aug 2018. doi: 10.1093/bioinformatics/bty766.

357 X.-Y. Liu, D. Bodaghi, Q. Xue, X. Zheng, and J.-X. Wang. Asynchronous parallel reinforcement
358 learning for optimizing propulsive performance in fin ray control. *Engineering with Computers*, 41
359 (4):2201–2218, 2025.

360 J. Metzcar, Y. Wang, R. Heiland, and P. Macklin. A review of cell-based computational modeling in
361 cancer biology. *JCO Clinical Cancer Informatics*, pages 1–13, Dec. 2019. ISSN 2473-4276. doi:
362 10.1200/cci.18.00069.

363 V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu.
364 Asynchronous methods for deep reinforcement learning. In *International conference on machine
365 learning*, pages 1928–1937. PmlR, 2016.

366 V. Noël, M. Ruscone, R. Shuttleworth, and C. K. Macnamara. Physimess - a new physicell addon
367 for extracellular matrix modelling. *Gigabyte*, 2024, Oct. 2024. ISSN 2709-4715. doi: 10.46471/
368 gigabyte.136.

369 K. O’shea and R. Nash. An introduction to convolutional neural networks. *arXiv preprint*
370 *arXiv:1511.08458*, 2015.

371 M. Ponce-de Leon, A. Montagud, V. Noël, A. Meert, G. Pradas, E. Barillot, L. Calzone, and
372 A. Valencia. Physiboss 2.0: a sustainable integration of stochastic boolean and agent-based
373 modelling frameworks. *npj Systems Biology and Applications*, 9(1):54, 2023.

374 M. L. Puterman. *Markov decision processes: discrete stochastic dynamic programming*. John Wiley
375 & Sons, 2014.

376 Python Software Foundation. Extending and embedding the python interpreter. <https://docs.python.org/3/extending/embedding.html>, 2024. Accessed: 2024-02-07.

378 L. H. Rocha, B. Aguilar, M. Getz, I. Shmulevich, and P. Macklin. A multiscale model of immune
379 surveillance in micrometastases gives insights on cancer patient digital twins. *npj Systems Biology*
380 *and Applications*, 10(1), Dec. 2024. ISSN 2056-7189. doi: 10.1038/s41540-024-00472-z.

381 M. Ruscone, A. Checcoli, R. Heiland, E. Barillot, P. Macklin, L. Calzone, and V. Noël. Building
382 multiscale models with physiboss, an agent-based modeling tool. *Briefings in Bioinformatics*, 25
383 (6), Sept. 2024. ISSN 1477-4054. doi: 10.1093/bib/bbae509.

384 M. Towers, A. Kwiatkowski, J. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão, A. Kallinteris,
385 M. Krimmel, A. KG, R. Perez-Vicente, A. Pierré, S. Schulhoff, J. J. Tai, H. Tan, and O. G. Younis.
386 Gymnasium: A standard interface for reinforcement learning environments, 2024.

387 G. van Rossum. Extending and embedding the python interpreter. *CWI Report CS-R9527*, May 1995.
388 URL <https://gvanrossum.github.io/Publications.html>.

389 C. J. Watkins and P. Dayan. Q-learning. *Machine learning*, 8:279–292, 1992.

390 A. E. Zade, S. S. Haghighi, and M. Soltani. Reinforcement learning for optimal scheduling of
391 glioblastoma treatment with temozolomide. *Computer methods and programs in biomedicine*, 193:
392 105443, 2020.

393 A Generation of Initial Conditions

394 **Rectangle (training layout).** Each cell type is placed uniformly in a vertical strip of width 10% of
395 the x -axis, with non-overlapping strip positions sampled from $[0.1, 0.2]$, $[0.3, 0.5]$, $[0.6, 0.8]$ (shuffled
396 across types). This is a highly regular, low-entropy layout.

397 **Network-field (held-out test layout).** White Gaussian noise $\eta(\mathbf{x}) \sim \mathcal{N}(0, 1)$ is convolved with a
398 Gaussian kernel G_σ ($\sigma = \ell_c/\sqrt{2}$, $\ell_c = 45 \mu\text{m}$) to produce a correlated field F . An additional local
399 noise component $\tilde{\xi}$ filtered at width $\ell_c/3$ is added with small weight α to introduce local heterogeneity.
400 The field is normalised to $[0, 1]$ and cells are sampled from positions with $\hat{F} > 0.55$, with probability
401 proportional to $\hat{F}$, producing irregular, clustered configurations that are structurally distinct from the
402 training rectangle.

403 B Training algorithm and setup

404 To address our control objective maximizing cancer-cell eradication while minimizing drug adminis-
405 tration we employ Soft Actor-Critic (SAC) [Haarnoja et al., 2018], a state-of-the-art RL algorithm
406 for continuous control. SAC is well suited to our setting due to its high sample efficiency, and as a
407 deep RL method it accommodates different observation types, such as image-based representations,
408 through Convolutional Neural Networks (CNNs) [O’shea and Nash, 2015]. A key challenge is the
409 computational cost of the simulator; to alleviate this bottleneck we adopt an asynchronous parallel
410 training framework [Mnih et al., 2016, Liu et al., 2025]. A GPU-resident learner updates policy and
411 value networks asynchronously while 13 CPU-resident environment processes generate experience in
412 parallel.

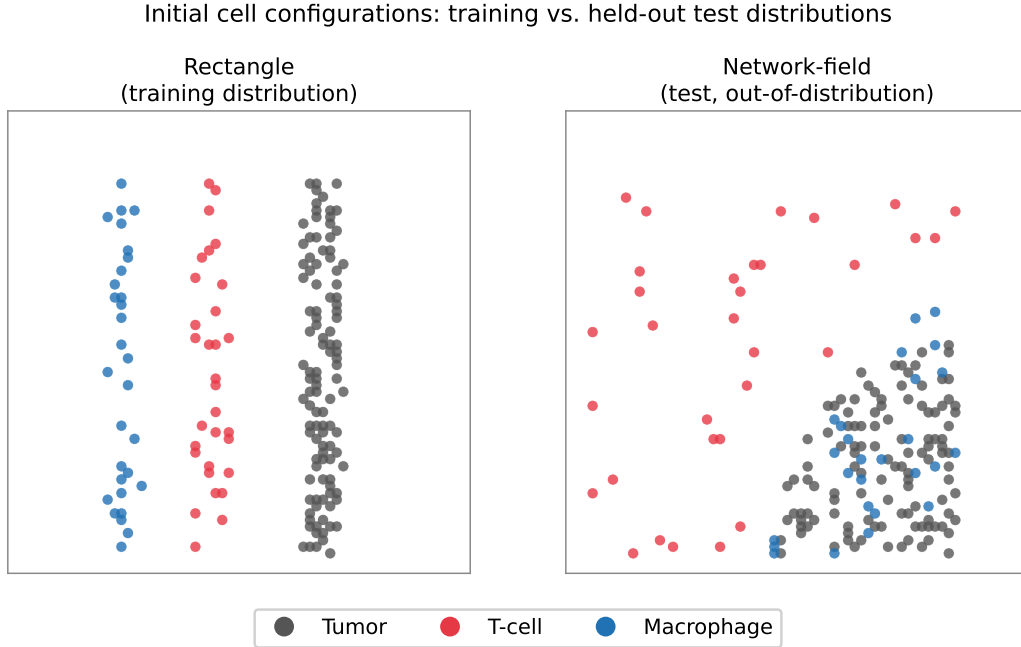


Figure 5: Representative examples of the training (rectangle) and held-out test (network-field) spatial layouts.

Compute. Each run trains a single observation mode for one seed to 10^5 agent decisions (6×10^5 simulator steps) in approximately 5 hours of wall-clock time on a single workstation (one GPU-resident learner with 13 CPU environment workers). The asynchronous design decouples the GPU learner from the comparatively slow PHYSICELL simulator, so throughput is limited by the CPU environment workers rather than by gradient updates; this is what makes the full observation-space sweep (nine modes $\times$ 5–7 seeds) tractable on a single machine.

Action repeat. Each action selected by the agent is held fixed for 6 consecutive simulator steps (*action repeat* of 6) before the next decision. This serves two purposes. First, it stabilises learning: with a 15-minute decision interval on a slow, indirect biological process (drug injection $\rightarrow$ macrophage repolarisation $\rightarrow$ gradient shift $\rightarrow$ T-cell migration $\rightarrow$ tumor death), consecutive raw simulator steps are highly correlated and a fresh decision at every step injects high-frequency jitter into the drug field without changing the underlying dynamics; committing to an action for several steps lets its effect materialise before it is revised, which empirically yields smoother dose trajectories and lower-variance value targets. Second, it improves simulator throughput per decision: the 10^5 agent decisions per run correspond to 6×10^5 simulator steps of experience, so the effective interaction budget (and all step axes in Figures 2–4) is 6×10^5 . Action repeat is a standard device for temporally extended control and is well motivated here by the slow timescale of the controlled process relative to the simulator step. The specific value $AR=6$ is selected by a sensitivity sweep over $AR \in \{1, \dots, 8\}$ (Appendix B.4, Figure 6): the action-smoothness cost drops steeply and plateaus by $AR \approx 6$, and the composite control objective reaches its plateau over $AR=5-8$, so $AR=6$ is the smallest repeat that attains both.

434 B.1 Asynchronous SAC

Algorithm 1 Asynchronous SAC with Parallel Environments [Liu et al., 2025]

Require: Config d_{arg} , buffer size $|\mathcal{B}|$, LRs, γ , τ

- 1: Init shared queues actor_queue, sample_queue; stop signal stop_event
- 2: **Actor Process (CPU):** init $N_{\text{env}} = 13$ envs, local π_{θ_a}
- 3: **while** not stop_event **do**
- 4: Receive θ from actor_queue if available
- 5: Act: random if $t < T_{\text{start}}$, else π_{θ_a}
- 6: Step envs; push (s, a, r, s', d) to sample_queue
- 7: **end while**
- 8: **Learner Process (GPU):** init π_{θ} , $Q_{\phi_{1,2}}$, targets, $\mathcal{D}$
- 9: **while** total samples $< T$ **do**
- 10: Drain sample_queue into $\mathcal{D}$
- 11: **for** N_{loops} gradient steps **do**
- 12: Sample minibatch; compute SAC target y ; update Q , π , α ; soft-update targets
- 13: **end for**
- 14: Push updated θ to actor_queue
- 15: **end while**

435 B.2 Neural Architecture

436 **Scalar modes** utilize a Multi-Layer Perceptron (MLP):

$$\text{FC}(256) \xrightarrow{\text{Mish}} \text{FC}(256)$$

437 **Image modes** utilize an IMPALA-style CNN [Espeholt et al., 2018]. The feature extraction and
 438 projection pipeline is defined as:

$$\begin{aligned} &\text{Conv}(n_c \rightarrow 32, k=8 \times 8, s=4) \xrightarrow{\text{Mish}} \\ &\text{Conv}(32 \rightarrow 64, k=4 \times 4, s=2) \xrightarrow{\text{Mish}} \\ &\text{Conv}(64 \rightarrow 64, k=3 \times 3, s=1) \xrightarrow{\text{Mish}} \text{Flatten} \rightarrow \text{FC}(256) \rightarrow \text{FC}(256) \end{aligned}$$

439 B.3 Hyperparameters

Table 2: SAC hyperparameters (TME v2 experiments).

Parameter	Symbol	Value
Agent decisions	T	1×10^5
Action repeat	–	6
Effective simulator steps	$6T$	6×10^5
Replay buffer size	$ \mathcal{B} $	2×10^5
Batch size	B	832
Warm-up steps	T_{start}	5,000
Policy / critic LR	η_{π}, η_Q	3×10^{-4}
Discount factor	γ	0.99
Target smoothing	τ	0.005
Reward weights	$w_{\text{cell}}, w_{\text{dose}}$	0.3, 2.0
Entropy autotuning	–	enabled
Parallel envs	N_{env}	13
Test frequency	–	every 4 episodes

440 B.4 Choice of action repeat

441 We select the action-repeat value with a dedicated sensitivity sweep, run independently of the main
 442 observation-space experiments. We sweep action_repeat $\in \{1, \dots, 8\}$ and, for each value, measure

three end-of-run quantities aggregated over repeated rollouts (with 95% bootstrap CIs): tumor reduction (control efficacy), an action-smoothness cost $\sum_t \|a_t - a_{t-1}\|^2$ (policy jitter; lower is better), and a composite objective combining tumor reduction, dose, and smoothness (Figure 6).

Two effects drive the choice. (i) *Smoothness*: the action-smoothness cost falls steeply and monotonically with action repeat from ≈ 110 at AR= 1 to ≈ 17 at AR= 6 and then flattens, so most of the stabilisation benefit is already captured by AR= 6 and larger values add little. This matches the physical timescale of the task: with a 15-minute decision interval on a slow immune process, re-deciding every simulator step injects high-frequency drug-field jitter that does not change the underlying dynamics. (ii) *Composite objective*: the objective is worst at AR= 1 (-0.88) and rises to a plateau over AR= 5–8, with AR= 6 at -0.07 ; within that plateau the values are within each other’s confidence intervals, so AR= 6 is chosen as the smallest repeat that reaches the objective plateau while sitting at the knee of the smoothness curve. This makes action repeat a deliberate, evidence-backed choice rather than an incidental hyperparameter.

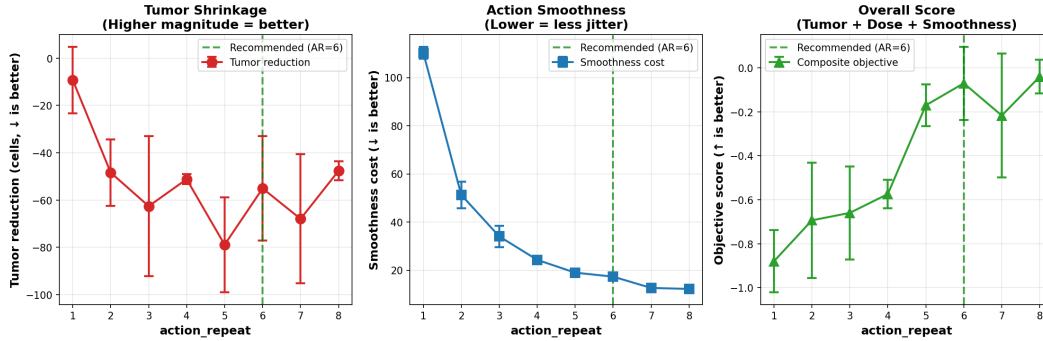


Figure 6: Action-repeat sensitivity sweep (independent of the observation-space experiments; error bars are 95% bootstrap CIs). **Left:** tumor reduction (more negative = more cells removed). **Centre:** action-smoothness cost drops sharply and plateaus by AR $\approx$ 6. **Right:** composite objective (higher is better) rises to a plateau over AR= 5–8; AR= 6 (dashed) is the smallest repeat on that plateau and at the knee of the smoothness curve.

C Full observation-mode comparison

Table 3 reports all nine observation modes evaluated, including the m1m2 scalar variants omitted from the curated main-text Table 1. Train and test returns are end-of-training EWMA-50 means with the standard deviation across seeds; train = rectangle, test = network-field.

Table 3: All observation modes (seeds averaged; train = rectangle, test = network-field). Modes are ordered by held-out test return.

ID	Mode	n	Train $_{\mu}$	Train $_{\sigma}$	Test $_{\mu}$	Test $_{\sigma}$
I2m	img_mc_cells_substrates_m1m2	5	+40.8	8.5	+32.5	9.4
I1	img_mc_cells	6	+30.5	8.7	+32.4	10.5
I2	img_mc_cells_substrates	7	+42.5	6.6	+30.4	7.4
I1m	img_mc_cells_m1m2	5	+35.0	13.1	+26.0	12.4
S5m	spatial_scalars_cells_spatial_no_scalars_substrates_m1m2	5	+53.6	34.2	+9.5	9.6
S3sm	spatial_scalars_cells_substrates_m1m2	5	+38.3	27.8	+2.6	12.1
S3m	spatial_scalars_cells_m1m2	6	+45.7	26.8	+2.6	9.7
S3s	spatial_scalars_cells_substrates	5	+42.9	18.8	−0.0	9.1
RAND	random policy (baseline)	5	−18.4	4.0	−25.3	3.0
POMDP	scalars_macrophages	5	−20.4	8.3	−31.8	2.3

D Bootstrap confidence intervals

Because each mode is trained with only $n = 5-7$ seeds, the across-seed standard deviations reported in Tables 1 and 3 are themselves noisy estimates of dispersion. To quantify uncertainty on the *mean* return more robustly, we bootstrap over seeds: for each mode we resample the seeds with replacement $B = 10,000$ times, recompute the mean end-of-training return of each resample, and report the 2.5th–97.5th percentiles of that bootstrap distribution as a 95% confidence interval on the mean (Table 4). The same procedure applied per step yields the shaded bands in Figures 7 and 8.

The bootstrap intervals sharpen the main-text claims. On the held-out network-field, the three image modes have test-return CIs that lie entirely above +24, whereas every spatial-scalar mode has a test-return CI that *contains zero* (e.g. S3s $[-6.2, +9.1]$), so their generalization is not statistically distinguishable from a non-learning policy. The image and spatial-scalar CIs do not overlap, confirming that the generalization gap is significant and not an artefact of seed selection. The scalars_macrophages POMDP baseline has a tight, strongly negative CI $[-33.4, -29.4]$, reflecting consistent failure rather than robustness and it does not overlap the random policy’s test CI $[-28.2, -22.8]$ from the favourable side, so the macrophage-only observation performs *no better than, and plausibly worse than, random dosing*.

Table 4: Mean end-of-training return with a 95% bootstrap confidence interval on the mean ($B = 10,000$ seed resamples; train = rectangle, test = network-field). Ordered by test mean.

ID	Mode	n	Train $_{\mu}$ (95% CI)	Test $_{\mu}$ (95% CI)
I2m	img_mc_cells_substrates_m1m2	5	+40.8 [+33.0, +47.3]	+32.5 [+24.4, +40.6]
I1	img_mc_cells	6	+30.5 [+23.1, +36.7]	+32.4 [+24.2, +40.6]
I2	img_mc_cells_substrates	7	+42.5 [+37.4, +47.2]	+30.4 [+24.5, +36.3]
I1m	img_mc_cells_m1m2	5	+35.0 [+22.7, +45.6]	+26.0 [+13.5, +38.5]
S5m	spatial_scalars_cells_spatial_no_scalars_substrates_m1m2	5	+53.6 [+25.0, +85.1]	+9.5 [+0.4, +17.6]
S3sm	spatial_scalars_cells_substrates_m1m2	5	+38.3 [+20.1, +65.9]	+2.6 [−6.2, +13.4]
S3m	spatial_scalars_cells_m1m2	6	+45.7 [+24.9, +67.3]	+2.6 [−5.8, +10.9]
S3s	spatial_scalars_cells_substrates	5	+42.9 [+24.7, +58.8]	−0.0 [−6.2, +9.1]
RAND	random policy (baseline)	5	−18.4 [−21.9, −14.9]	−25.3 [−28.2, −22.8]
POMDP	scalars_macrophages	5	−20.4 [−28.0, −13.8]	−31.8 [−33.4, −29.4]

E Example Episode Comparisons

Figure 9 shows a single matched test episode comparing an image mode (img_mc_cells_substrates), a spatial-scalar mode (spatial_scalars_cells_substrates), and the POMDP baseline (scalars_macrophages) step by step. To guarantee an exact comparison, all three agents are rolled out (deterministically, using the policy mean action) from the *identical* held-out network-field initial condition the same 124-tumor-cell layout replayed from file so the only difference between the three trajectories is the observation mode.

The episode cleanly separates the three regimes. img_mc_cells_substrates drives the tumor to near-eradication (3 cells remaining, +223 cumulative return); it spends the most drug (4.5 total dose) but places it precisely enough visibly dosing in sustained, targeted bursts (Figure 9, bottom) to convert dose into effective macrophage repolarisation and tumor clearance. spatial_scalars_cells_substrates achieves partial control (70 cells, +30 return, 2.8 dose): it suppresses growth early but plateaus and cannot finish, lacking the spatial substrate feedback needed to target the final push. scalars_macrophages (POMDP) barely acts (0.9 total dose) and the tumor *grows* past its starting size (130 cells, −6.4 return): seeing only macrophage-polarisation counts, it cannot localise where to inject, so it under-doses and fails consistent with its aggregate performance at or below the random baseline.

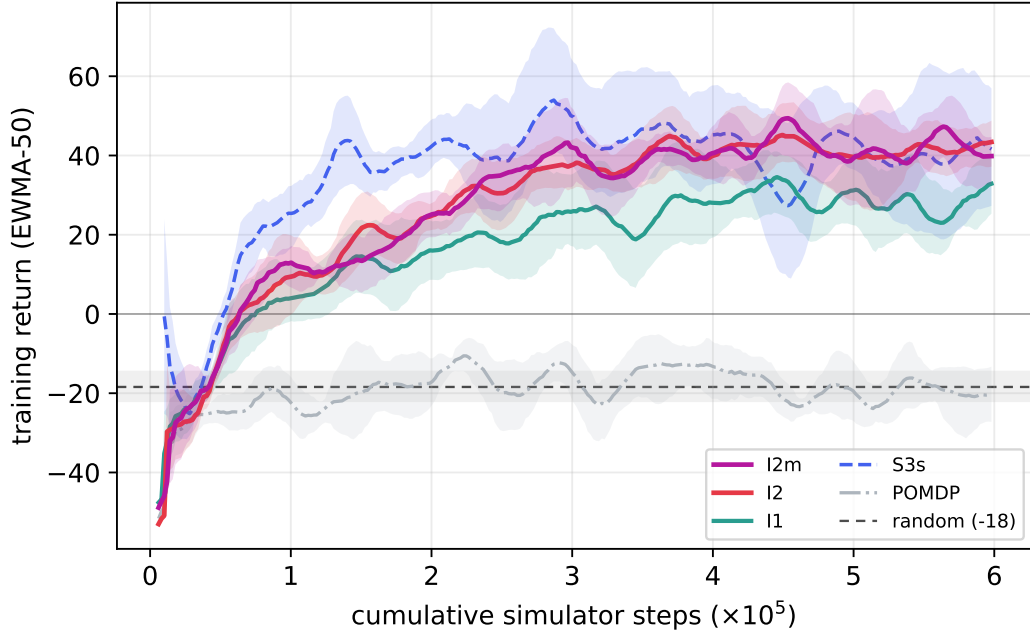


Figure 7: Training return on rectangle (EWMA-50, mean with 95% bootstrap CI band over seeds) vs. cumulative simulator steps. Curated main modes only.

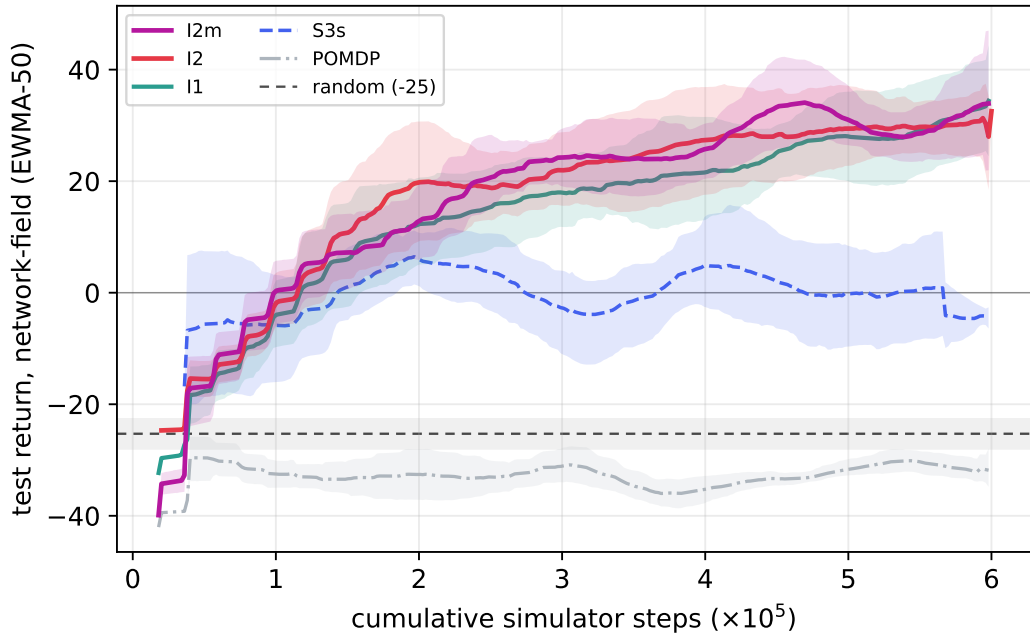


Figure 8: Held-out test return on network-field (EWMA-50, mean with 95% bootstrap CI band over seeds) vs. cumulative simulator steps. The image modes' bands separate from the spatial-scalar summary (S3s), whose band hugs zero, and from the POMDP baseline.

Matched network-field episode (identical IC, 124 tumor cells)

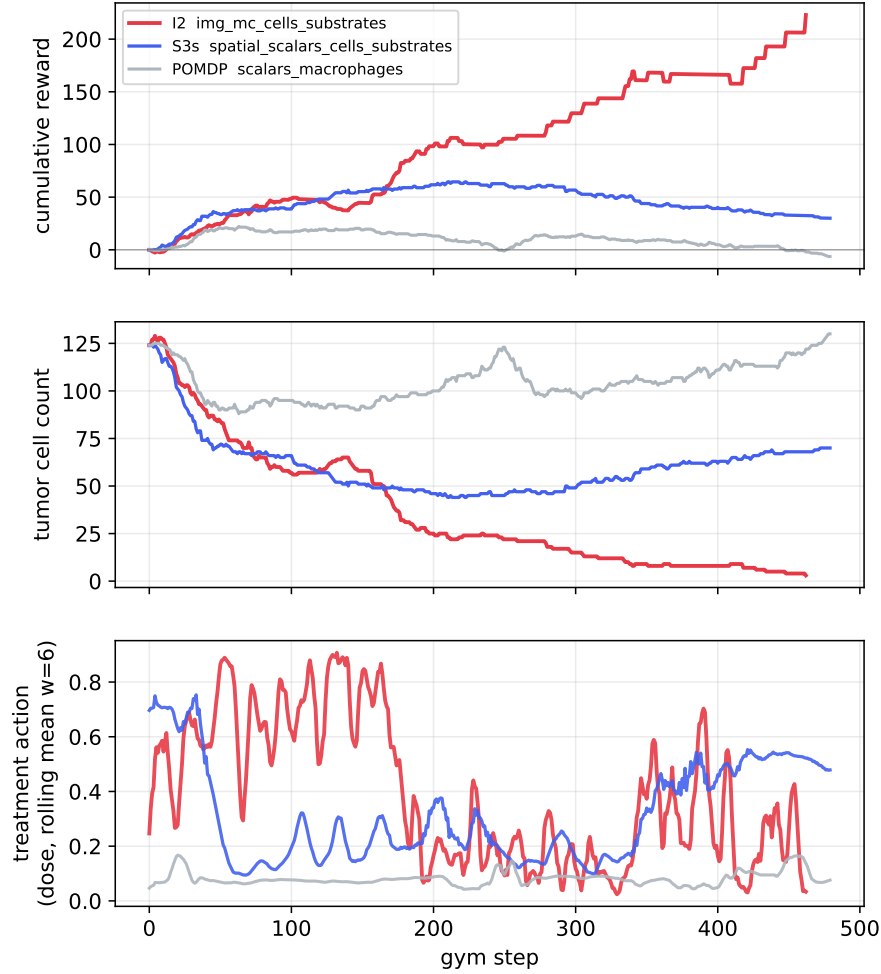


Figure 9: **Matched network-field episode (identical initial condition, 124 tumor cells)**. Cumulative reward (top), tumor cell count (middle), and the dose action (bottom, rolling mean over 6 steps matching the action-repeat horizon used in training; the raw per-step signal is noisy and unreadable at full resolution) for `img_mc_cells_substrates` (red), `spatial_scalars_cells_substrates` (blue), and the `scalars_macrophages` POMDP baseline (grey), all replayed from the same IC. `img_mc_cells_substrates` reaches near-eradication (3 cells, +223 return, 4.5 dose); `spatial_scalars_cells_substrates` achieves partial control (70 cells, +30 return, 2.8 dose); the POMDP baseline under-doses and the tumor grows (130 cells, -6.4 return, 0.9 dose). Only the image mode sustains a high, decisive dose action long enough to drive the tumor to near-eradication, whereas `spatial_scalars_cells_substrates` oscillates and eventually lets the tumor regrow. Note that the physical dose delivered to the simulation each step is the concentration action (bottom) multiplied by the voxel volume swept by the independently-controlled injection radius action, so the dose action alone does not linearly predict the physical dose spent.